SMART LOGISTICS DECISION SYSTEM

ML-Based Risk Intelligence & Cost Optimization Platform

PRESENTED BY

Meet Dinesh Patil 23BDS0083

Anish Mukul Ringe 23BDS0082

PROBLEM & SYSTEM VISION

Industry Problem

Delivery delays significantly **increase operational costs** and fuel consumption.

SLA penalties and refunds **reduce business reliability** and margins.

Reactive manual planning is **inefficient** for modern scale.

Need: **Predictive logistics intelligence** to anticipate risks.

PROBLEM & SYSTEM VISION

Project Objective

- System **predicts delay probability** for incoming shipments.
- Classifies risk levels: **Low, Medium, High, Critical**.
- Estimates potential **financial exposure** (SLA penalties & refunds).
- **Optimizes ETA** dynamically based on real-time factors.
- Deploys insights via an interactive **real-time dashboard**.

Problem Framing

CENTRAL QUESTION

"Will this specific delivery be delayed?"

BINARY CLASSIFICATION TASK

0

ON-TIME DELIVERY

Target State

1

DELAYED DELIVERY

Risk Event

Next Phase

Now we move to **data processing** and **model development**.

Dataset Overview

Operational Features: Traffic status, asset utilization, and waiting times.

Environmental Context: Precipitation levels and temporal features (hour of day, peak vs. off-peak).

Transactional Data: Order value and shipment priority classification.

Target Variable: Binary classification label (0 = On-time, 1 = Delayed).

DATA PROCESSING & MODEL DEVELOPMENT

Data Preprocessing

1

Missing Value Handling

Imputation strategies applied to ensure data completeness without bias.

2

Duplicate Check

Removal of redundant records to prevent data leakage and bias.

3

Categorical Encoding

One-Hot encoding for nominal variables to numerical format.

4

Standard Scaler

Normalization (mean=0, std=1) for Logistic Regression stability.

5

Feature Order Preservation

Maintaining strict column indexing for consistent inference.

Pipeline ensures data consistency from training to real-time deployment

Model Selection

Chosen Model: Logistic Regression

Selected for robustness, speed, and transparency in risk scoring.

Interpretable coefficients to explain risk factors (e.g., Rain impact).

Direct **Probability Output** (0-1) essential for risk engines.

Deployment-Ready with ultra-low latency for real-time dashboards.

Industry-Proven baseline for binary classification tasks.

Training Strategy

Train-Test Split

80/20 standard split to preserve unseen data for final evaluation.

5-Fold Cross Validation

Iterative training on different data subsets to ensure model robustness.

Hyperparameter Tuning

Optimizing C (Inverse Regularization Strength) to balance bias and variance.

L2 Regularization

Applied to prevent overfitting by penalizing large coefficients.

Cross-Validation Process

Ensuring model generalizability across 5 iterations

Fold 1	Test	Train	Train	Train	Train
Fold 2	Train	Test	Train	Train	Train
Fold 3	Train	Train	Test	Train	Train
Fold 4	Train	Train	Train	Test	Train
Fold 5	Train	Train	Train	Train	Test

Training Data Validation Data

Phase Complete

PHASE 2 COMPLETED

Model Training

PHASE 3 NEXT

Model Evaluation

"After training the model, we **evaluated its performance**."

MODEL EVALUATION & RISK ENGINE

Model Performance

ROC-AUC SCORE

0.798

Indicates strong discriminatory power

PRECISION

0.985

ACCURACY

0.775

RECALL

0.610

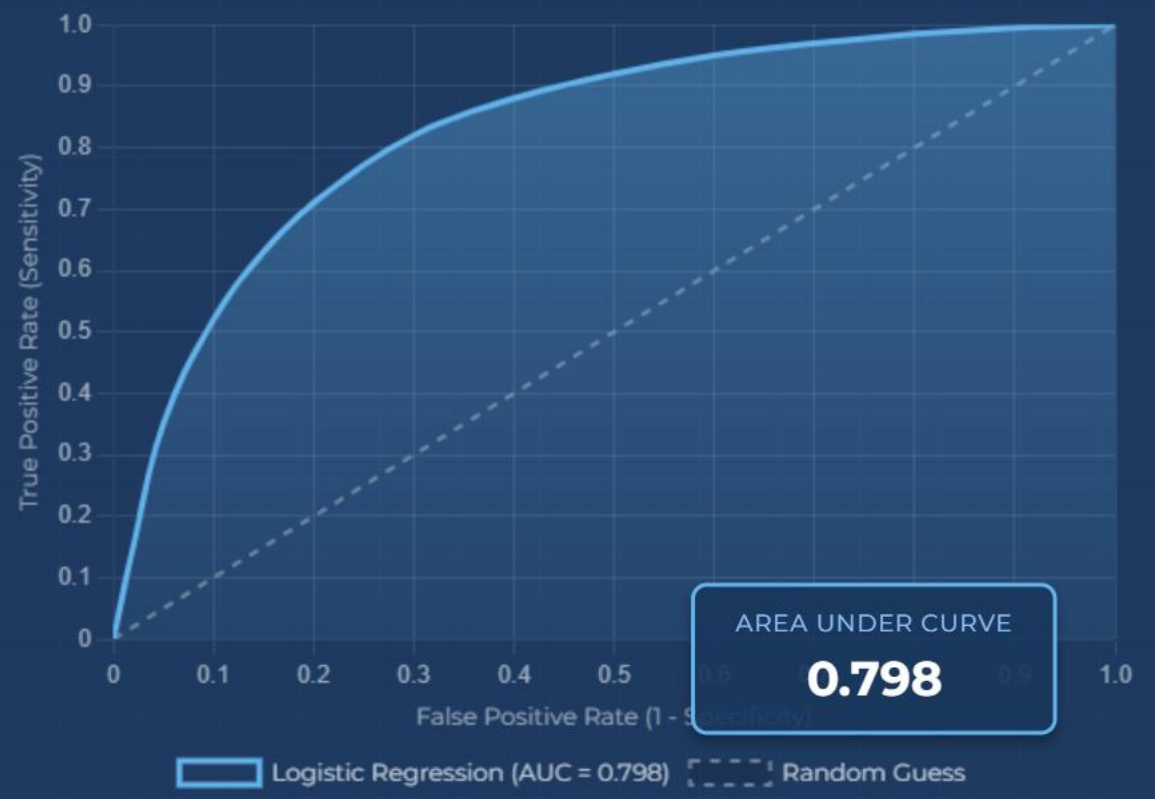
F1 SCORE

0.754

Deployment Decision:

Threshold Adjusted to 0.60

Optimized to reduce false positives in operational planning.



Why ROC-AUC?

- **Threshold-independent evaluation**: Assesses performance across all possible decision boundaries, not just a single cutoff.
- Measures **ranking capability**: How well the model distinguishes between delayed and on-time classes.
- Robust under **class imbalance**: Provides a reliable metric even when one class (e.g., delayed) is rare.
- Provides a **comprehensive performance view** by summarizing the trade-off between sensitivity and specificity.

MODEL EVALUATION & RISK ENGINE

Threshold Calibration

Decision boundary shifted from default **0.5** to adjusted **0.6**.

Primary Purpose: Significantly **reduce false positives** to prevent alarm fatigue.

Ensures better **operational trade-off** by prioritizing high-confidence alerts.

Distinction: ROC-AUC measures *model potential*, while Threshold dictates *deployment action*.

MODEL EVALUATION & RISK ENGINE

Risk Classification Logic

PROBABILITY THRESHOLDS		OPERATIONAL ACTIONS
< 0.40	LOW RISK	Proceed with **Normal Operations**
0.40 – 0.70	MEDIUM RISK	**Monitor Closely** for status changes
0.70 – 0.88	HIGH RISK	Trigger **Reroute / Notify** Stakeholders
≥ 0.88	CRITICAL RISK	Immediate **Reroute + Redistribute** Load

Phase Transition

"After determining **risk levels**, we extend into **ETA optimization** and **cost modeling**."

NEXT: OPTIMIZATION ENGINE →

Baseline ETA Calculation

$$\text{Baseline ETA} = \text{Op. Base Time} + (\text{Delay Prob.} \times \text{Op. Base Time})$$

RESULT INTERPRETATION

A data-driven estimated arrival time that accounts for the inherent risk of delay, moving beyond static averages.

OPERATIONAL BASE TIME

Historical average transit time for the specific route and carrier under normal conditions.

RISK ADJUSTMENT

Additional buffer time calculated by the ML model's probability output (0.0 – 1.0) multiplied by the trip duration.

This baseline provides the foundation before traffic-specific optimization is applied.

Optimized ETA

$$ETA_{opt} = Base + p \times Base \times (1 - T)$$

TRAFFIC FACTOR (T)

A real-time coefficient [0, 1] derived from live traffic data. It represents the potential efficiency gain available through dynamic rerouting.

BOUNDED IMPROVEMENT

Mathematical constraints ensure the optimized ETA never exceeds physical limitations, preventing unrealistic delivery promises.

RISK APPLICATION

This advanced optimization is selectively triggered only for **High** and **Critical** risk shipments to recover lost time.

*Where **p** = Delay Probability and **Base** = Operational Base Time.*

Financial Impact Engine

MATHEMATICAL CORE

DELAY ESTIMATION MODEL

Expected Delay Hours

Delay Probability × Operational Base Time

FINANCIAL EXPOSURE

Expected Loss ($)

Delay Probability × Total Delay Cost

COST STRUCTURE COMPONENTS

SLA Penalty
8% – 15% of contract value per violation

Refund Risk
5% × Order Value × Probability

Extra Shipping Cost
20% premium for emergency routing

Express Penalty
Variable based on carrier & distance

ETA OPTIMIZATION & DEPLOYMENT

Batch Shipment Intelligence

STEP 01

Upload **CSV Dataset** containing multiple shipment records.

STEP 02

Predict **Delay Risk** for the entire shipment portfolio.

STEP 03

Calculate **Expected Loss** per shipment (Prob × Cost).

STEP 04

Compute **Priority Score** to rank critical interventions.

STEP 05

Visualize total **Portfolio Exposure** and risk distribution.

STEP 06

Download **Comprehensive Report** for operational action.

Streamlit Deployment & Architecture

REAL-TIME DASHBOARD FEATURES

Shipment Input Interface
Direct parameter entry for instant analysis

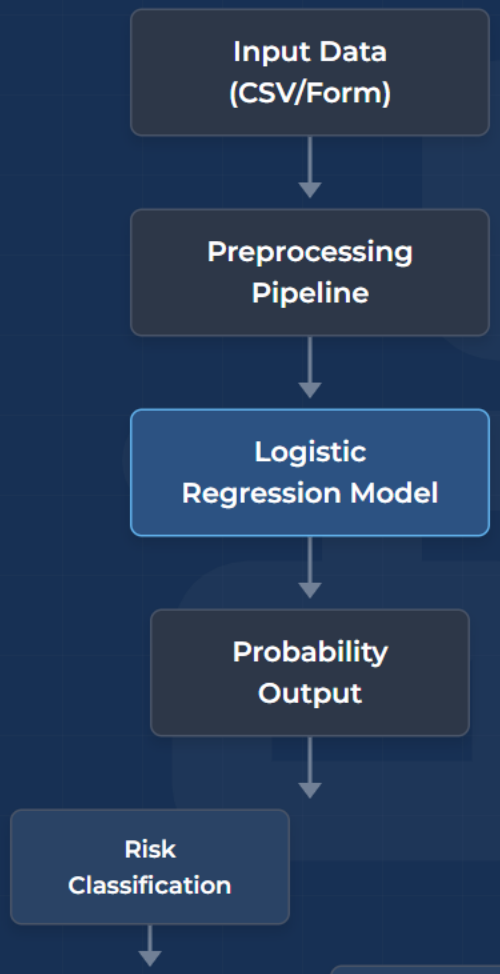
Delay Probability & Risk Level
Immediate visual classification (Low to Critical)

Dynamic ETA Display
Side-by-side Baseline vs. Optimized ETA

Financial Exposure
Real-time cost calculation based on risk

Customer Notification
Automated alerts for high-risk shipments

END-TO-END SYSTEM ARCHITECTURE



Project Conclusion

From Raw Data to Strategic Action

We have successfully engineered a complete ML lifecycle that transforms operational uncertainty into a **Deployment-Ready Logistics Intelligence Platform**.

ROBUST ML PIPELINE

Rigorous preprocessing and 5-fold cross-validation yielded a stable model with **0.798** AUC and high precision.

CALIBRATED RISK ENGINE

Threshold optimization (0.60) balances operational trade-offs, reducing false alarms while capturing critical delays.

OPERATIONAL VALUE

Dynamic ETA adjustments and financial impact modeling provide direct, actionable insights for cost reduction.

SYSTEM READY FOR EVALUATION